

BRENHAM, TX, USA

WMO: 722553

Lat: 30.219N

Lon: 96.374W

Elev: 94

StdP: 100.20

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) 0.0	(d) -8.5	(e) 1.8	(f) 4.4	(g) -6.3	(h) 2.2	(i) 6.6	(j) 11.2	(k) 13.0	(l) 9.9	(m) 12.9	(n) 2.1	(o) 350	(p) 0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.3	(c) 37.4	(d) 23.7	(e) 36.3	(f) 24.0	(g) 35.2	(h) 24.3	(i) 26.8	(j) 31.3	(k) 26.3	(l) 30.9	(m) 25.9	(n) 30.5	(o) 3.2	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.6	27.7	25.3	20.7	27.3	24.9	20.2	27.3	84.1	31.1	82.1	31.3	80.4	30.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	8.0	7.1	-4.9	39.1	2.5	1.5	-6.7	40.2	-8.1	41.0	-9.5	41.9	-11.3	42.9
			-5.6	27.3	2.6	0.7	-7.4	27.8	-8.9	28.2	-10.4	28.6	-12.2	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	11.1	13.2	17.0	20.3	24.6	28.2	29.1	29.6	26.8	21.8	16.4	12.3
	DBStd	7.70	5.51	5.77	4.75	3.83	3.13	1.84	1.78	1.80	2.61	4.30	5.20	5.37
	HDD10.0	136	53	31	6	1	0	0	0	0	0	0	9	37
	HDD18.3	816	233	159	82	24	2	0	0	0	0	18	96	201
	CDD10.0	4111	86	121	224	309	452	544	592	607	503	366	201	107
	CDD18.3	1748	8	16	41	82	196	294	333	349	253	126	38	13
	CDH23.3	18511	33	72	224	603	1831	3397	3973	4282	2611	1156	264	65
	CDH26.7	8599	2	8	32	128	697	1697	2046	2315	1218	405	48	4
Wind	WSAvg	2.7	3.0	3.3	3.2	3.3	3.1	2.5	2.3	2.1	1.8	2.2	2.6	2.9
Precipitation	PrecAvg	1107	80	78	71	83	121	119	57	72	123	112	106	85
	PrecMax	1641	201	209	221	212	332	398	199	284	395	445	270	285
	PrecMin	541	5	6	6	5	2	0	2	1	6	9	14	14
	PrecStd	269	50	48	47	53	75	87	48	62	83	101	68	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	27.9	29.1	31.3	34.8	37.7	37.9	38.7	37.3	34.2	30.0	27.4
		MCWB	18.8	19.5	19.8	21.5	23.5	23.2	23.5	23.7	22.3	23.6	22.4	20.4
	2%	DB	24.0	25.6	27.5	29.4	32.9	36.2	36.9	37.6	35.7	32.5	27.9	25.1
		MCWB	17.7	19.0	19.6	21.1	23.4	23.3	23.9	23.9	23.4	23.5	21.1	19.9
	5%	DB	22.3	23.7	26.0	27.9	32.1	34.8	35.9	36.5	34.0	30.9	26.4	23.3
		MCWB	17.6	18.5	19.1	20.6	23.2	23.8	24.3	24.4	23.7	23.0	19.8	19.6
	10%	DB	20.7	22.2	24.1	27.1	30.9	33.5	34.8	35.6	32.7	28.9	24.7	22.1
		MCWB	16.8	18.1	18.6	19.9	22.9	24.0	24.4	24.6	23.5	21.5	19.8	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.8	22.5	24.6	26.2	27.1	27.3	27.4	26.9	26.0	24.3	22.9
		MCDB	23.3	24.4	25.3	27.9	29.9	31.0	32.3	32.0	30.2	30.0	27.1	25.0
	2%	WB	20.1	21.0	21.8	23.4	25.4	26.4	26.6	26.7	26.1	25.2	23.2	21.7
		MCDB	22.6	23.4	24.7	26.9	29.4	30.4	31.4	31.7	29.9	28.9	26.0	24.0
	5%	WB	19.1	20.1	21.0	22.6	24.7	25.8	26.1	26.2	25.6	24.6	22.1	20.6
		MCDB	21.5	22.7	23.6	26.0	28.8	30.1	30.9	31.4	29.7	28.3	24.9	22.7
	10%	WB	17.4	19.0	20.2	21.8	24.1	25.4	25.7	25.7	25.1	23.8	21.0	19.0
		MCDB	19.0	21.5	23.0	25.0	28.2	29.8	30.5	30.9	29.4	27.4	23.4	21.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.7	11.1	11.3	10.5	10.6	10.4	11.3	11.0	12.2	11.7	11.6
		MCDBR	12.8	11.8	11.5	11.5	11.2	12.2	12.0	12.6	12.3	12.0	12.4	12.6
		MCWBR	7.7	6.9	5.6	5.2	3.6	3.0	3.0	3.0	3.4	5.0	6.8	8.6
	5% WB	MCDBR	10.2	9.8	9.3	9.2	9.3	9.8	10.4	11.2	10.2	10.3	10.2	11.3
		MCWBR	7.8	7.0	5.9	5.4	4.0	3.2	3.3	3.3	3.4	5.2	6.4	9.4
Clear-Sky Solar Irradiance	taub	0.330	0.347	0.368	0.419	0.469	0.448	0.469	0.458	0.428	0.377	0.370	0.329	
	taud	2.494	2.459	2.400	2.266	2.170	2.281	2.251	2.277	2.319	2.442	2.399	2.507	
	Ebn at Noon	903	917	917	876	831	843	824	831	845	872	846	886	
	Edh at Noon	90	102	115	136	151	135	138	133	124	102	99	85	
All-Sky Solar Radiation	RadAvg	2.89	3.26	4.40	5.33	5.80	6.34	6.40	6.05	5.11	4.28	3.19	2.55	
	RadStd	0.29	0.61	0.40	0.40	0.43	0.47	0.51	0.36	0.48	0.54	0.35	0.32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+87

Nomenclature: See separate page